

E²CNN: entity-type-enriched cascaded neural network for Chinese financial relation extraction

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Abstract Knowledge Graphs (KGs) are pivotal for effectively organizing and managing structured information across various applications. Financial KGs have been successfully employed in advancing applications such as audit, anti-fraud, and anti-money laundering. Despite their success, the construction of Chinese financial KGs has seen limited research due to the complex semantics. A significant challenge is the overlap triples problem, where entities feature in multiple relations within a sentence, hampering extraction accuracy – more than 39% of the triples in Chinese datasets exhibit the overlap triples. To address this, we propose the Entity-type-Enriched Cascaded Neural Network (E²CNN), leveraging special tokens for entity boundaries and types. E²CNN ensures consistency in entity types and excludes specific relations, mitigating overlap triple problems and enhancing relation extraction. Besides, we introduce the available Chinese financial dataset FinCORPUS.CN, annotated from annual reports of 2,000 companies, containing 48,389 entities and 23,368 triples. Experimental results on the DUE dataset and FinCORPUS.CN underscore E²CNN's superiority over state-of-the-art models.

Keywords financial knowledge graph, overlap triples, cascaded neural network, relation extraction

1 Introduction

Knowledge graphs have become a widely adopted standard for knowledge representation in the semantic web, where knowledge is encoded as a collection of “facts”. Typically, the facts are expressed as triples of the form (*subject*, *predicate*, *object*), where the subject and object are entities, and the predicate indicates a relation between them. In general, knowledge graphs can be applied to support various downstream tasks, such as question answering [1,2], data integration [3], and fact-checking [4].

Currently, significant efforts have been invested in

constructing a knowledge graph (KG) with a primary focus on named entity recognition [5,6] and relation extraction [7–9]. Several domain knowledge graphs have already been established, including those for food [10] and manufacturing [11]. In this research, we focus on constructing a Chinese financial knowledge graph. This resource holds potential applications in diverse anti-fraud services, such as consumer credit services [12], credit card applications [13], fraud detection, and identification [14]. Although some attempts have been made to build language-independent neural networks for knowledge graph construction from diverse language corpora [15], extracting relations from the Chinese financial corpus remains challenging due to its unique characteristics.

One obstacle is the presence of the same entity appearing in multiple relations within a single sentence. According to our statistics from the collected Chinese financial corpus, nearly 62.91% of entities exhibit this phenomenon, which is also known as the overlap triple problem [16,17]. In Fig. 1, we provide an example to illustrate the entity pair overlap (EPO) and single entity overlap (SEO) patterns of overlap triples. For instance, consider the sentence “Suzhou Jiangnan Film Vice President Zhang Hua has become a board member and investor of Yunhai Film.” Multiple relations exist between “Zhang Hua” and “Yunhai Film”. Furthermore, “Suzhou Jiangnan Film” serves as the subject in the triple (Suzhou Jiangnan Film, locate, Suzhou) and as the object in the triple (Zhang Hua, belong, Suzhou Jiangnan Film).

To tackle these issues, we employ a cascaded neural network. Initially, we detect a group of candidate subjects. Thereafter, for each subject, we proceed to extract all related objects via pre-defined relations, addressing the entity pair overlap. For single entity overlap, since extraction is structured for each relation, every subject can match its potential objects in different relations.

Besides, we discover that the relation types in the Chinese financial corpus exhibit a strong correlation with the entity types, and both the boundaries and types of entities offer a valuable opportunity. Such improvement materializes in a two-fold manner. **Consistency and exclusion in relation**